ASIAN INSTITUTE OF MANAGEMENT

Machine Learning 3

Convolutional Neural Networks

Sessions 2-3

MSDS 2026

17 April 2026

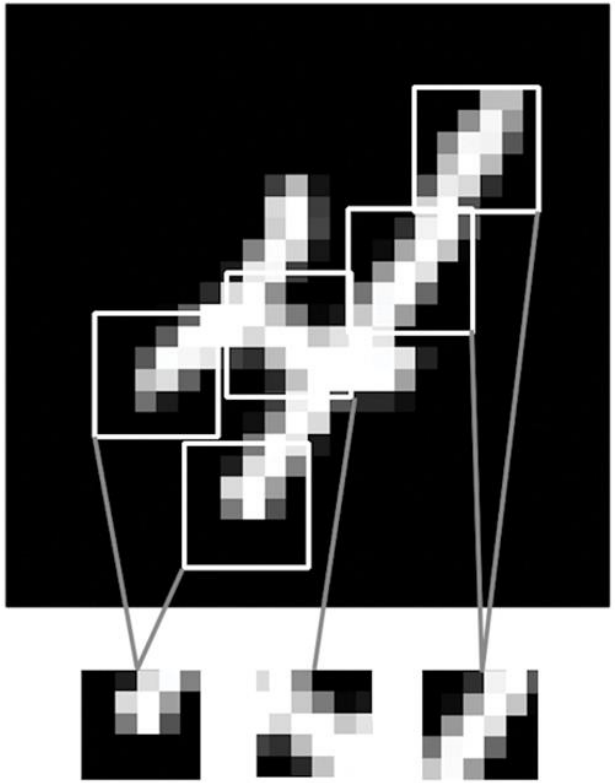
ASIAN INSTITUTE OF MANAGEMENT

Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNN)

a.k.a. convolutional networks (ConvNets)

- employs a mathematical operation called **convolution**, a specialized kind of linear operation
- learns local patterns



Chollet, F., & Watson, M. (2026). *Deep Learning with Python Third Edition.* Manning.

Convolutional Neural Networks

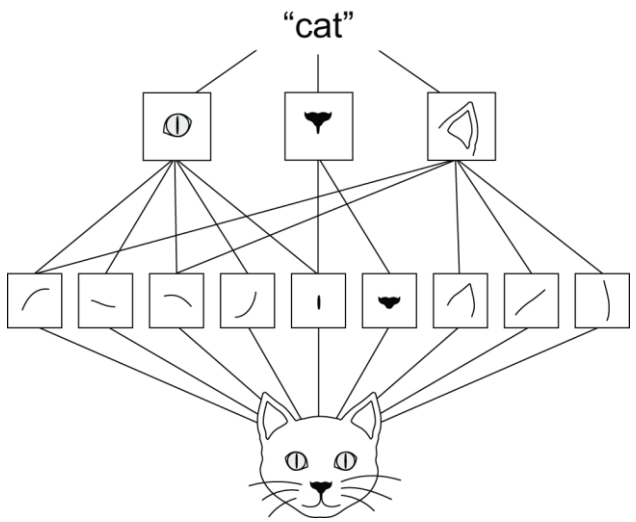
Properties

TRANSLATION INVARIANT

- **The patterns they learn are translation invariant**

SPATIAL HIERARCHIES

- **They can learn the spatial hierarchies of patterns**



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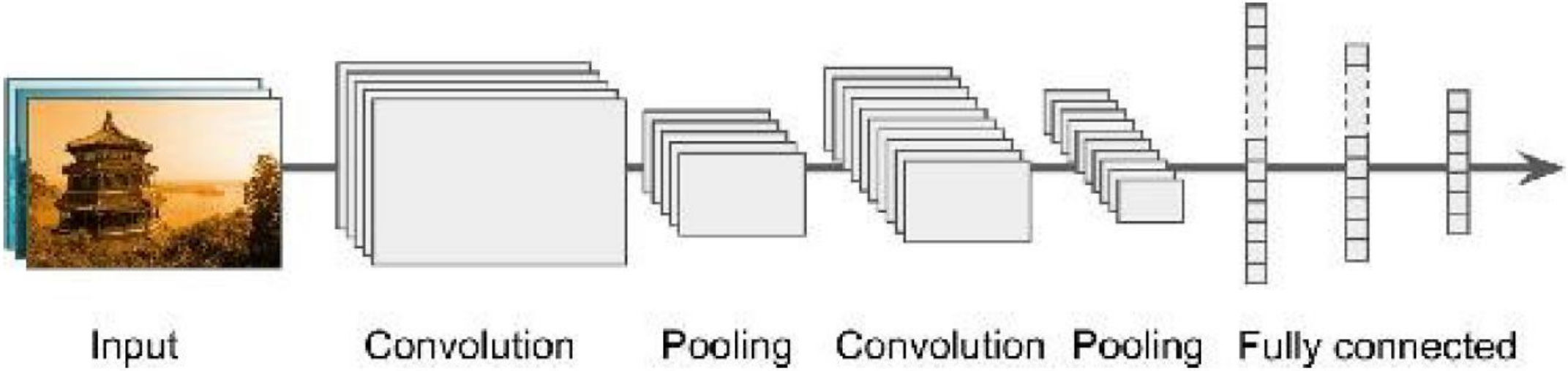
CNN Architecture

CNN Architecture

modularity-hierarchy-reuse formula



Convolution

Convolutional Neural Networks

Convolution Operation

- extracts patches from its **input feature map** and applies the same transformation to all these patches, producing an **output feature map**
- works by sliding the filters over the input feature map, stopping at every possible location and extracting the patch of surrounding features

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Convolution Operation

SPARSE-CONNECTIVITY

- accomplished by making the kernel smaller than the input

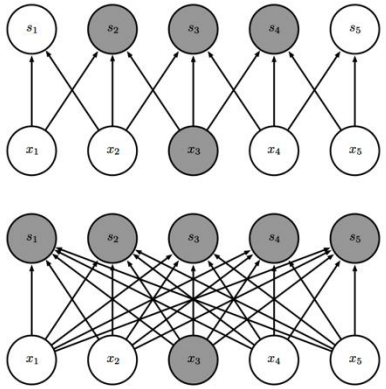

Figure 9.2: Sparse connectivity, viewed from below. We highlight one input unit, x_3, and highlight the output units in $\boldsymbol{s}$ that are affected by this unit. *(Top)*When $\boldsymbol{s}$ is formed by convolution with a kernel of width 3, only three outputs are affected by $\boldsymbol{x}$. *(Bottom)*When $\boldsymbol{s}$ is formed by matrix multiplication, connectivity is no longer sparse, so all the outputs are affected by x_3.

PARAMETER-SHARING

- using the same parameter for more than one function in a model

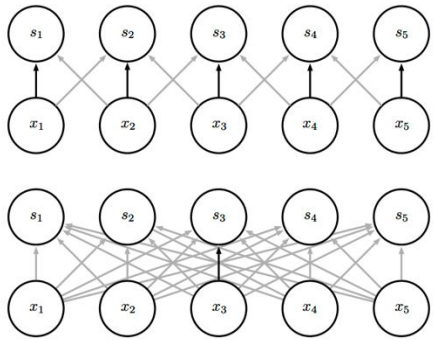

Figure 9.5: Parameter sharing. Black arrows indicate the connections that use a particular parameter in two different models. *(Top)*The black arrows indicate uses of the central element of a 3-element kernel in a convolutional model. Because of parameter sharing, this single parameter is used at all input locations. *(Bottom)*The single black arrow indicates the use of the central element of the weight matrix in a fully connected model. This model has no parameter sharing, so the parameter is used only once.

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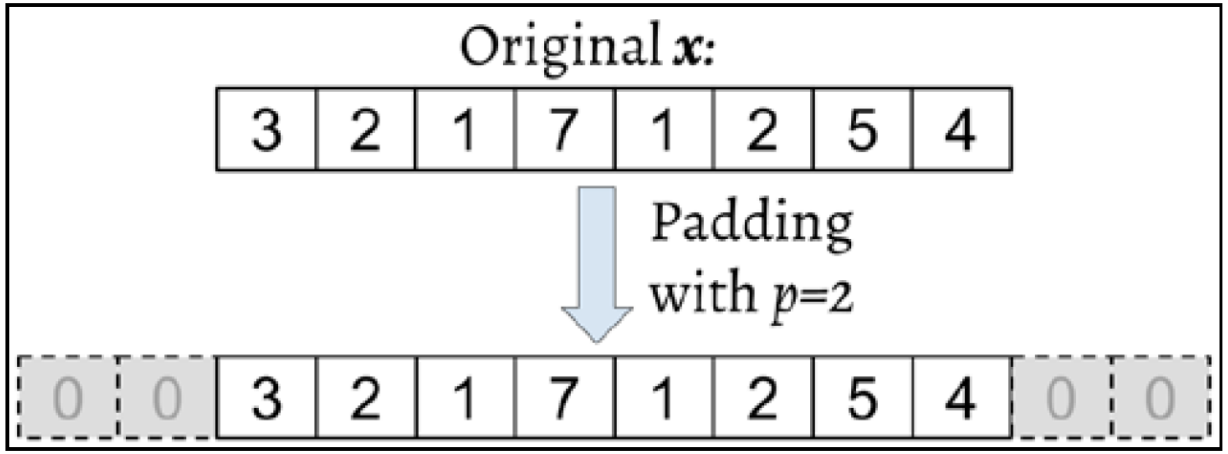
Convolutional Neural Networks

Convolution Operation

PADDING (*p*)

- to pad the edges of the inputs with new values and proceed as usual to retain the input shape



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Convolutional Neural Networks

Convolution Operation

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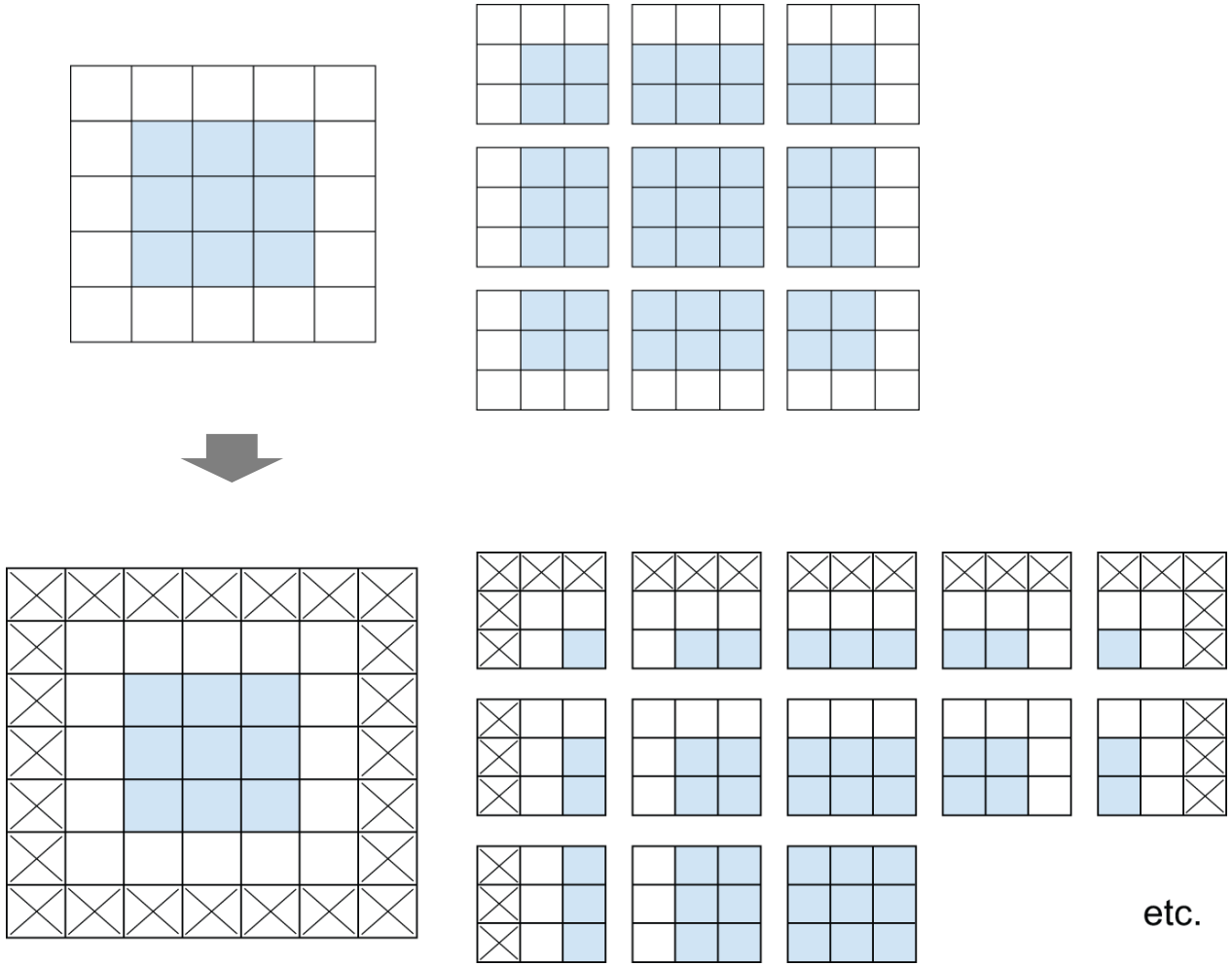

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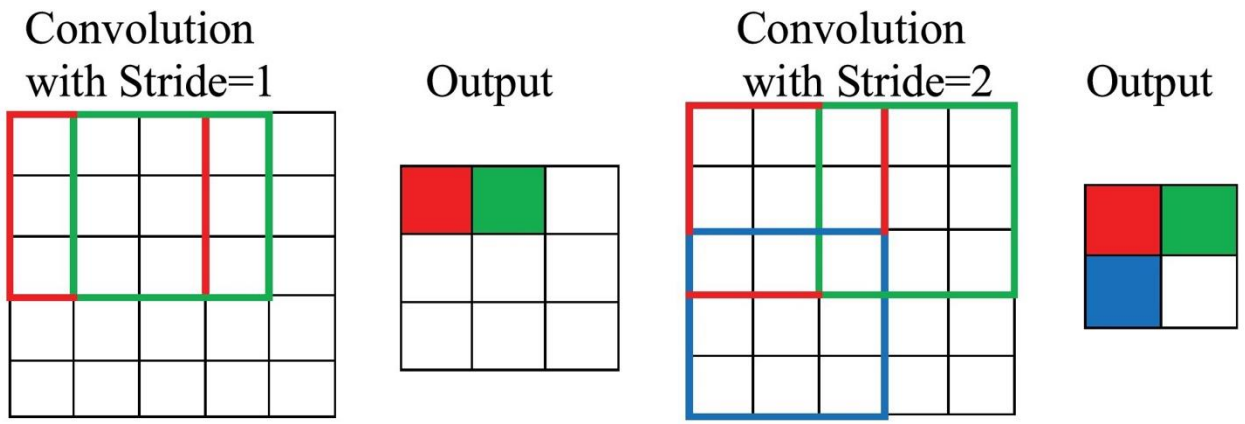
Convolution Operation

STRIDE (*s*)

- the distance between two successive windows



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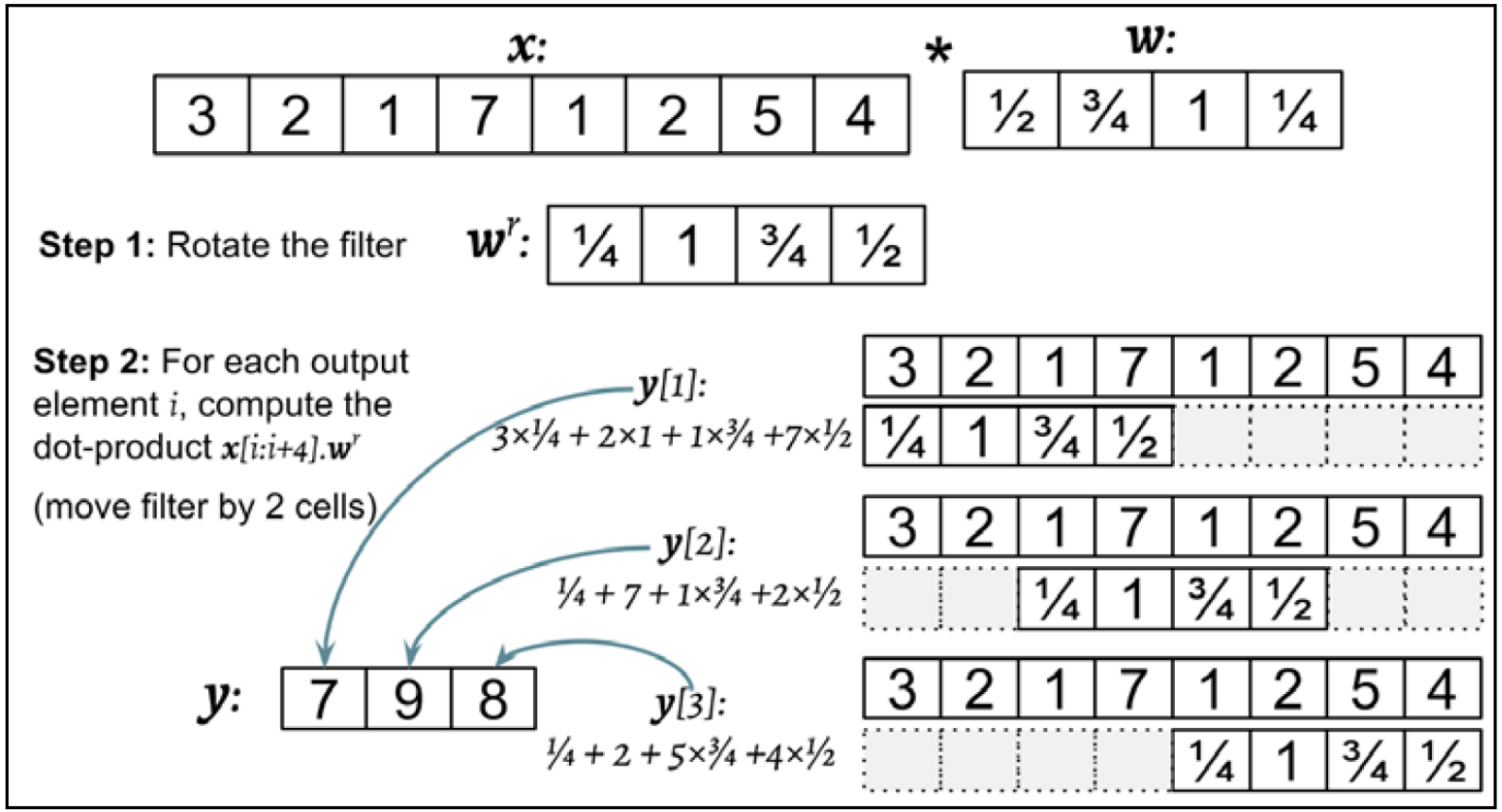
1-D Convolution Operation



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1-D Convolution Operation

OUTPUT SIZE

- This is determined by the total number of times we shift the filter *w* along the input vector.
- The input vector *x* has size *n* and the filter *w is of size m*.

$$o = floor(\frac{n + 2p - m}{s}) + 1$$

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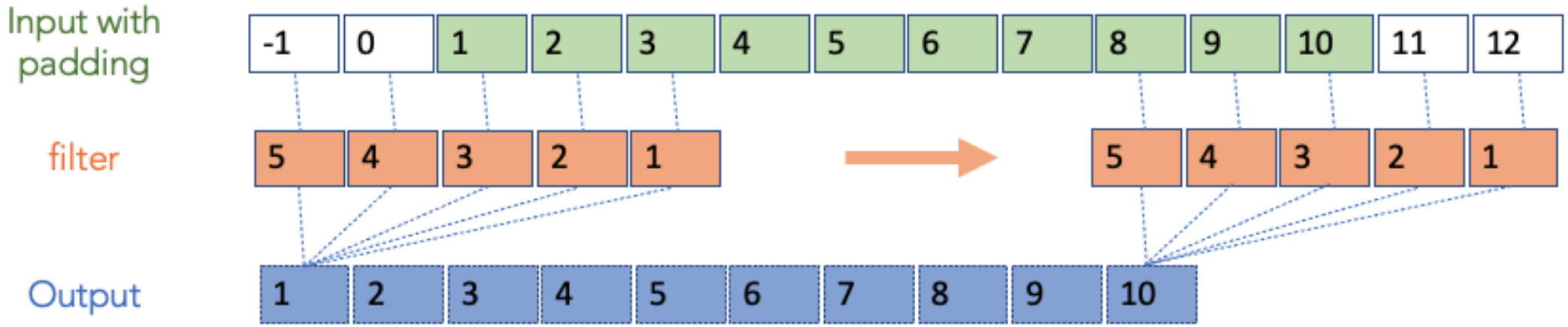
Convolutional Neural Netwo

$$o = floor(\frac{n + 2p - m}{s}) + 1$$

1-D Convolution Operation

OUTPUT SIZE



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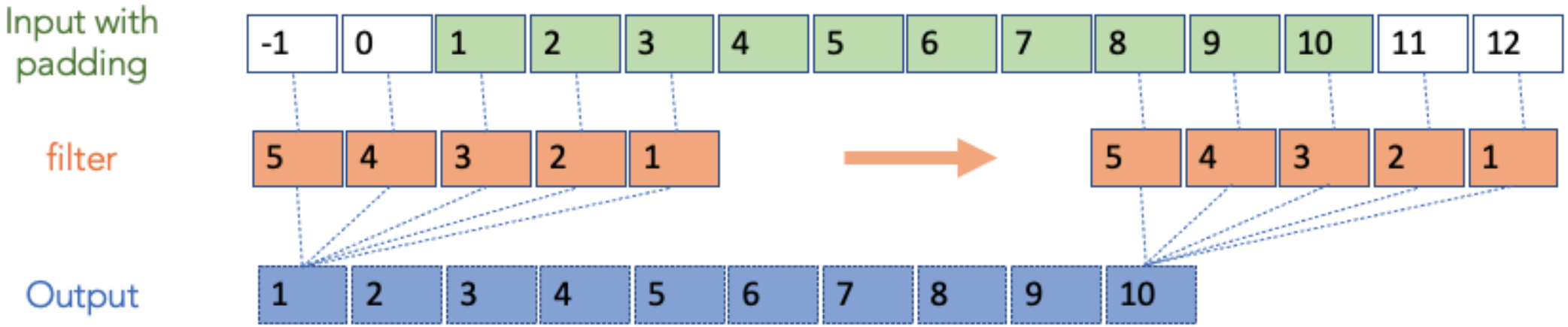
Convolutional Neural Netwo

$$o = floor(\frac{n + 2p - m}{s}) + 1$$

1-D Convolution Operation

OUTPUT SIZE



$$n = 10, m = 5, p = 2, s = 1, \Rightarrow o = floor(\frac{10 + 2 \times 2 - 5}{1}) + 1 = 10$$

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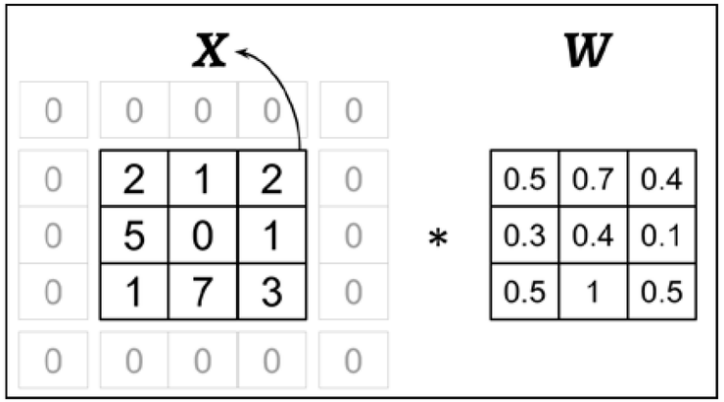
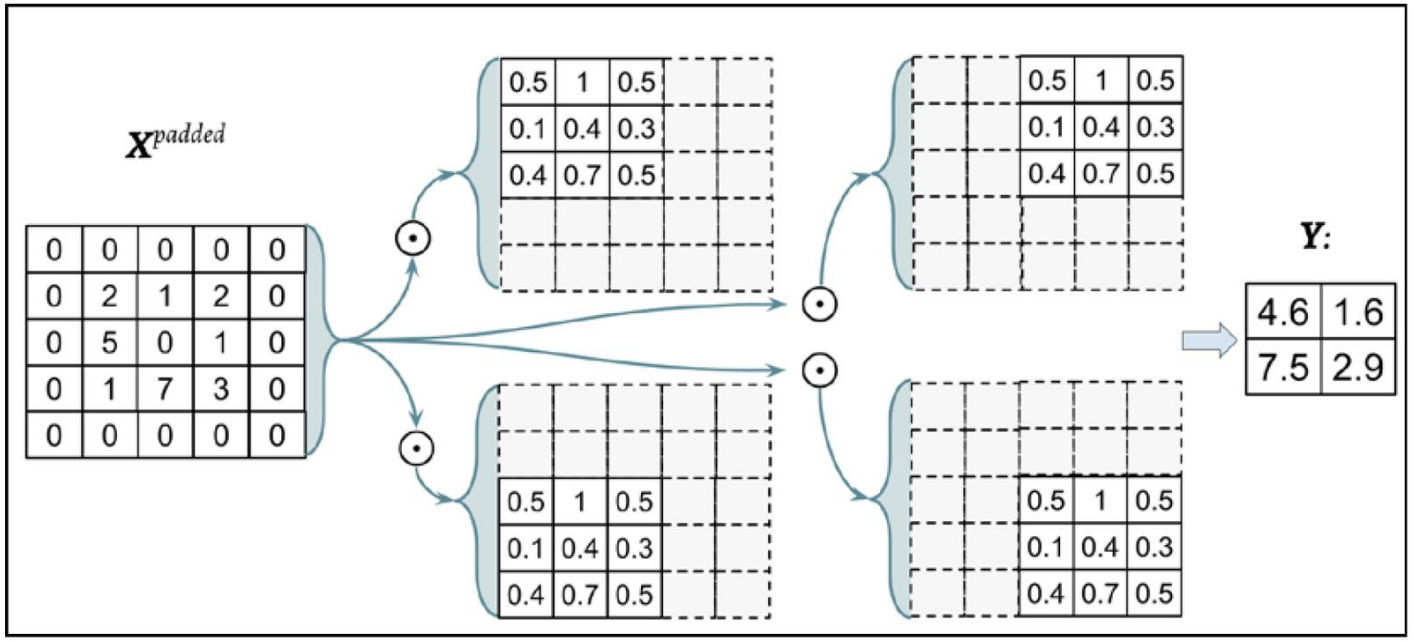
2-D Convolution Operation



$$W^r = \begin{bmatrix} 0.5 & 1 & 0.5 \\ 0.1 & 0.4 & 0.3 \\ 0.4 & 0.7 & 0.5 \end{bmatrix}$$



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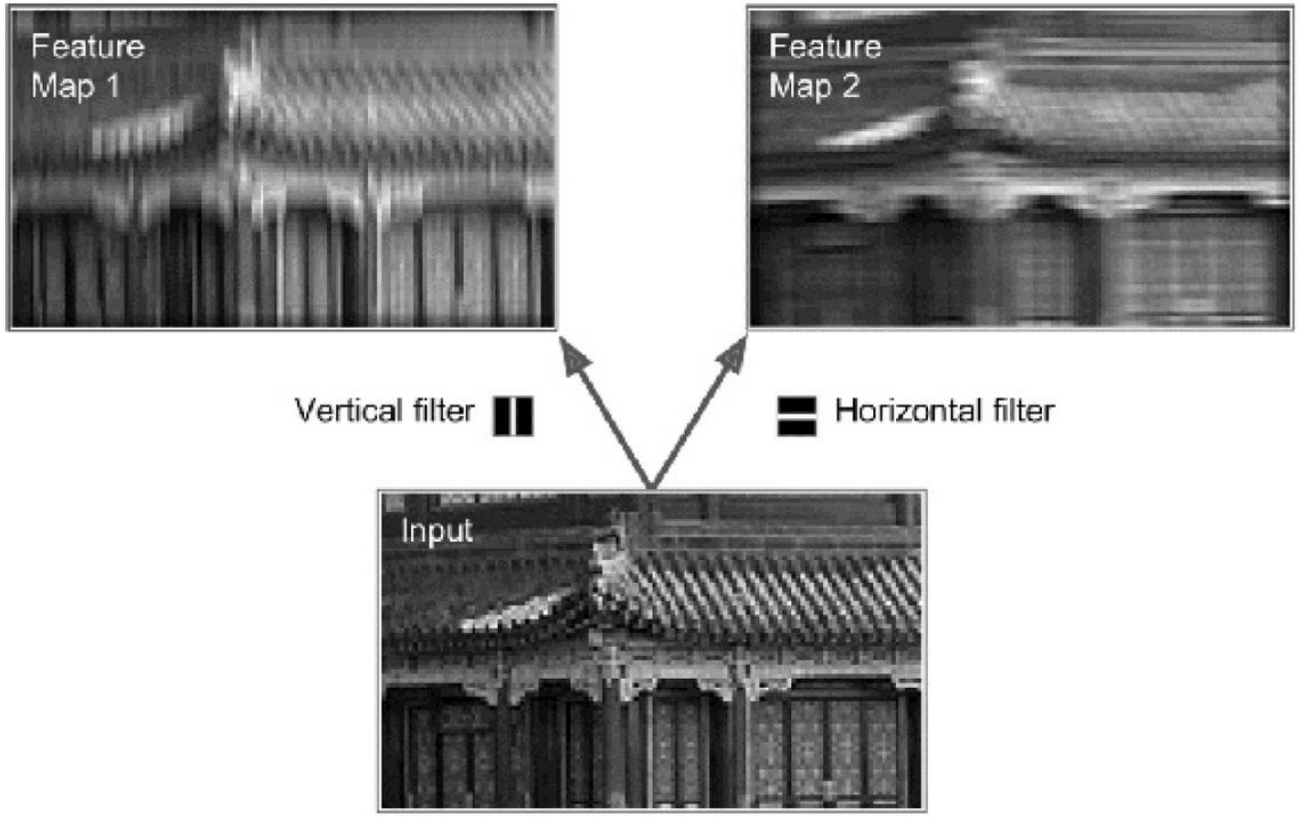
Convolutions as FILTERS



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Pooling Layers

Convolutional Neural Networks

Pooling Layer

replaces the output of the net at a certain location with a summary statistic of the nearby outputs

$$\left.\begin{array}{l} \boldsymbol{X}_1 = \begin{bmatrix} 10 & 255 & 125 & 0 & 170 & 100 \\ 70 & 255 & 105 & 25 & 25 & 70 \\ 255 & 0 & 150 & 0 & 10 & 10 \\ 0 & 255 & 10 & 10 & 150 & 20 \\ 70 & 15 & 200 & 100 & 95 & 0 \\ 35 & 25 & 100 & 20 & 0 & 60 \end{bmatrix} \\ \boldsymbol{X}_2 = \begin{bmatrix} 100 & 100 & 100 & 50 & 100 & 50 \\ 95 & 255 & 100 & 125 & 125 & 170 \\ 80 & 40 & 10 & 10 & 125 & 150 \\ 255 & 30 & 150 & 20 & 120 & 125 \\ 30 & 30 & 150 & 100 & 70 & 70 \\ 70 & 30 & 100 & 200 & 70 & 95 \end{bmatrix} \end{array}\right\} \xrightarrow{\text{max-pooling } P_{2\times 2}} \begin{bmatrix} 255 & 125 & 170 \\ 255 & 150 & 150 \\ 70 & 200 & 95 \end{bmatrix}$$

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cnn-1 Notebook

Convolutional Neural Networks

cnn-1 Notebook

EXPLORATION TASKS

- Task 1: Perform discrete convolutions in 1D
- Task 2: Apply a 2D convolution layer with four (4) 3x3 filters to a sample image
- Task 3: Explore the use of different operations and layers
- Task 4: Create an initial CNN model to classify handwritten digits

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Exercise 3: Handwritten Digit Classification

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e3-cnn-classification

INSTRUCTIONS

1. **Improve the initial model until you obtain your best model.** Consider tuning key gradient descent parameters, tuning the representational power of your model by chaning the number and/or type of layers or the size of the layers, or applying regularizations.
2. **Report the model performances** on the training, validation and test sets of your initial and best models. Describe the steps you undertook to improve the performance of your initial model.
3. Generate your **own set of handwritten number images** from 0 to 9. Test your best model on this set and report its performance.

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Reminders

Reminders

Next Session

Session 5 and 6 (April 24, 2026) – Convolutional Neural Network

Deliverables

Due this week

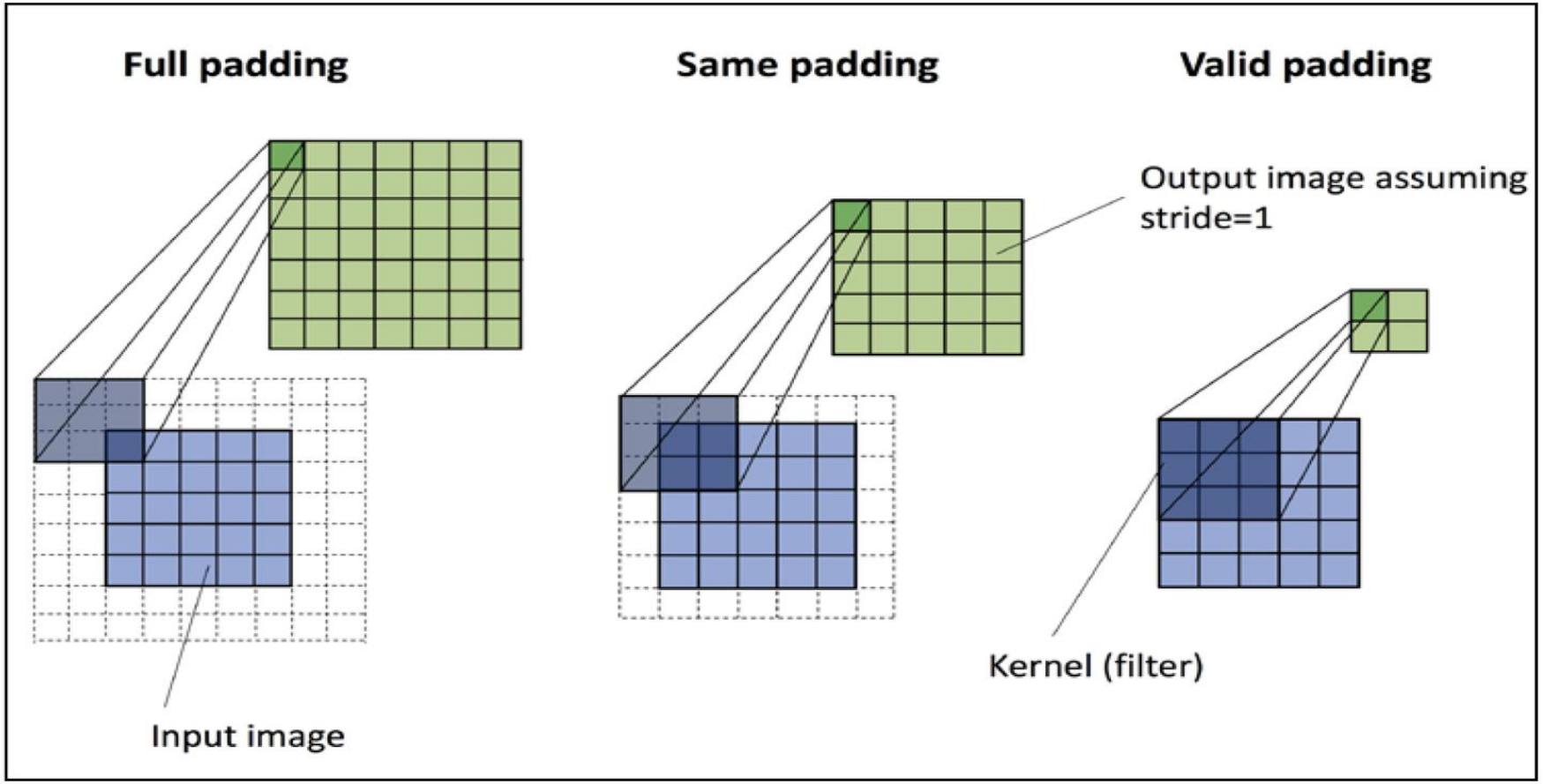
- R02 - Post-Class Reflection 2 (due EOD April 17, 2026)

Due next week

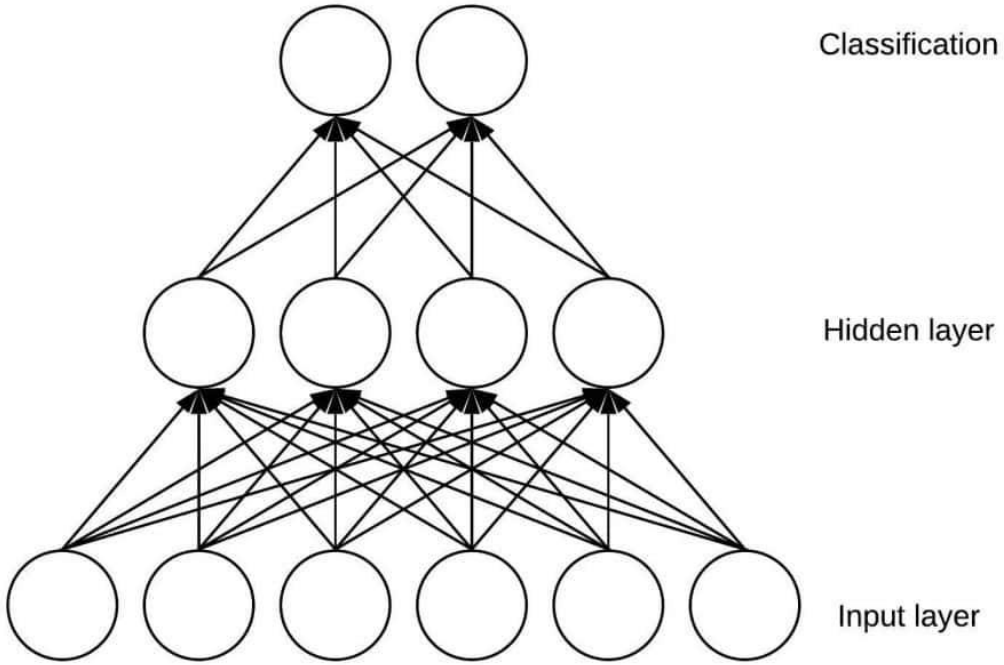
- E3 - Exercise 3 (due EOD April 24, 2026)

Illustrations

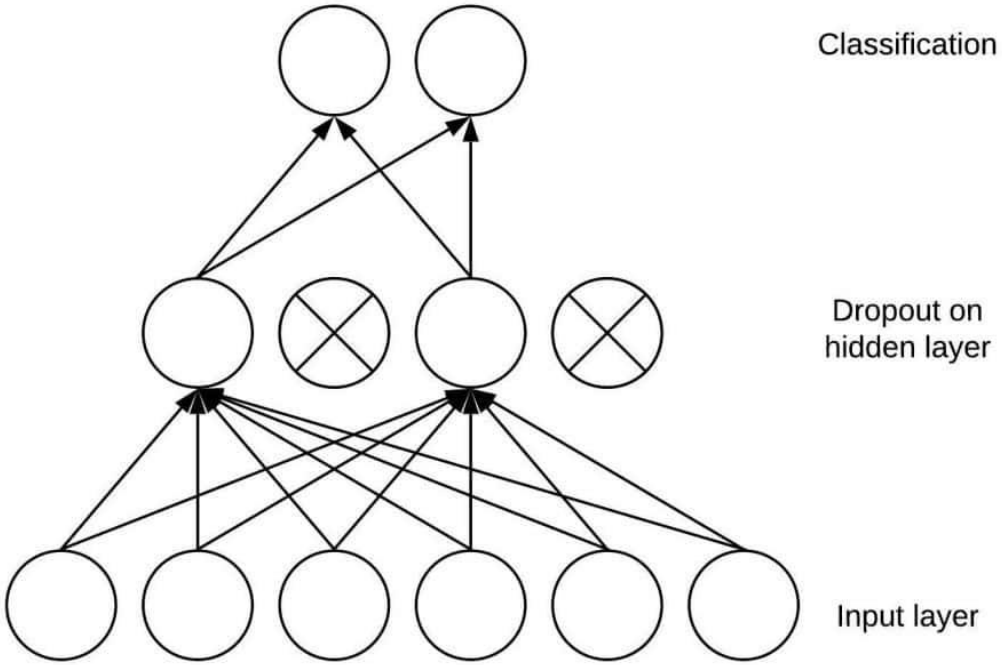
Padding


Full padding
Same padding
Valid padding
Output image assuming stride=1
Kernel (filter)
Input image

Dropout

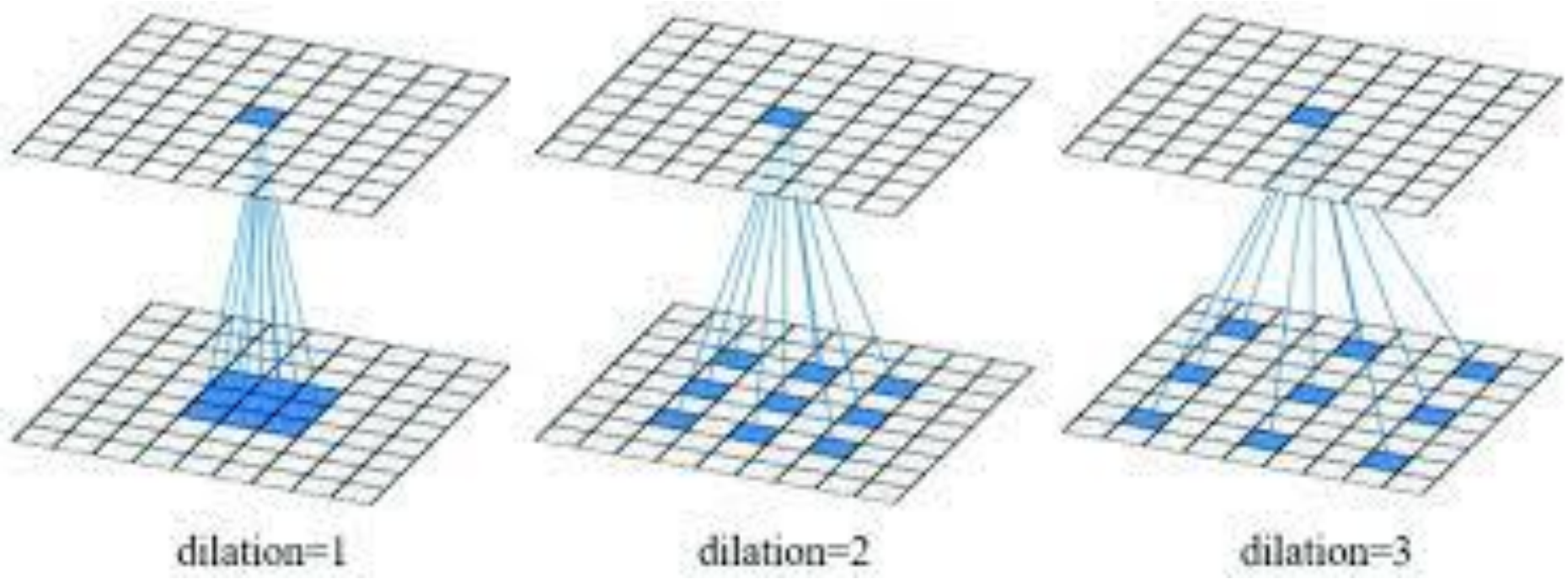


Without Dropout



With Dropout

Dilation



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Machine Learning 3

Fundamentals of Machine Learning

Session 2

MSDS 2026

10 April 2026

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